MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 6115A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex oscillator, integral cavity, mechanically tuned frequency range 5,100 to 5,900 MHz

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Ik	Ehk	Alt
Unit:	V	Vdc	Vdc	mAdc	Vdc	ft
	6.8	330	-350	35	50	10,000
	5.8	---	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Cathode:	Unipotential, oxide coated
Base:	Phenolic wafer (modified octal)
Weight:	1.5 ounces (approx)
Mounting position:	Any

TEST CONDITIONS: See notes 1 and 2

Parameter:	Ef	Ers	Er	F
	V	Vdc	Vdc	MHz
	6.3	300	-60 to -220	F2

FREQUENCY		
F	MHz	±
F1	5,100	0.3%
F2	5,500	0.3%
F3	5,900	0.3%

GENERAL:

Qualification - Required

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:43Z
Check the source to verify that this is the current version before use.

OD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
	High-frequency vibration	-		---	---	---	---
	Temperature coefficient	-		ΔF/ΔT	---	-0.10 +0.05	MHz/°C
	Frequency change	3		ΔF	---	15	MHz
	Shock	4	Test condition A, except 15 G	Po	---	10	%
	Quality conformance inspection, part 1	5					
	Cathode emission	-	Ef = 5.8 V	ΔIk/Ik	---	15	%
	Electrode current (cathode)	-		Ik	---	32	mAdc
	Power output	-	Ef = 5.8 V	Po	70	---	mW
	Reflector voltage	-	Ef = 5.8 V; F = F2	Er	-115	-175	Vdc
	Total reflector current	6,7		Ir	---	7.0	μAdc
	Reflector leakage current	6,7		Ir	---	5.0	μAdc
	Reflector gas current	6,7		Ir	---	2.0	μAdc
	Mechanical tuning range	-	High frequency, Er = -135 to -205 Vdc	F	5,900	---	MHz
			Low frequency, Er = -85 to -145 Vdc	F	---	5,100	MHz
	Quality conformance inspection, part 2						
	Heater current (nonoperating)	-		If	410	470	mA
	Insulation of electrodes	-	Tube cold; 300 Vdc	Rk-rs Rh-rs	2.0 2.0	--- ---	Meg Meg
	Heater-cathode leakage	-	Ehk = ±45 Vdc	Ihk	0	100	μAdc
	Electronic tuning range	-	Er/50% max Po	ΔF	30	---	MHz
	Quality conformance inspection, part 3						
	Life test	1	Group B	t	500	---	hrs
	Life-test end point:						
	Power output	-		Po	35	---	mW

:

The TUT shall be mounted in a socket as shown on figure 2 and shall be fixed firmly by means of clamps which make contact with the tube only at the flange adjacent to the base. The measurements on the tubes in an oscillating state shall be made with the tube coupled to the waveguide as shown on figure 2.

No warmup time is required in order to obtain the specified life from the tube. If extended life is desired, a heater warmup period of approximately 1 minute is recommended before the application of the resonator voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:43Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

3. The TUT shall be operated at a maximum power output for 10 minutes under voltages as specified in test conditions at the frequency specified herein. The TUT shall then be operated for 2 minutes with filament voltage only. Ten seconds after all voltages have been reapplied, the tube frequency shall not differ from the original frequency by more than the amount specified herein.

4. The TUT shall be shocked five times in each of the three planes. The change in power output after shock in any one plane shall not exceed the specified value.

5. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

6. After 2 minutes with all voltages applied, total reflector current shall not exceed the specified limits.

7. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 70, 80
DSA - ES

User activities:
Army - WC, ME
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3065)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:43Z
Check the source to verify that this is the current version before use.

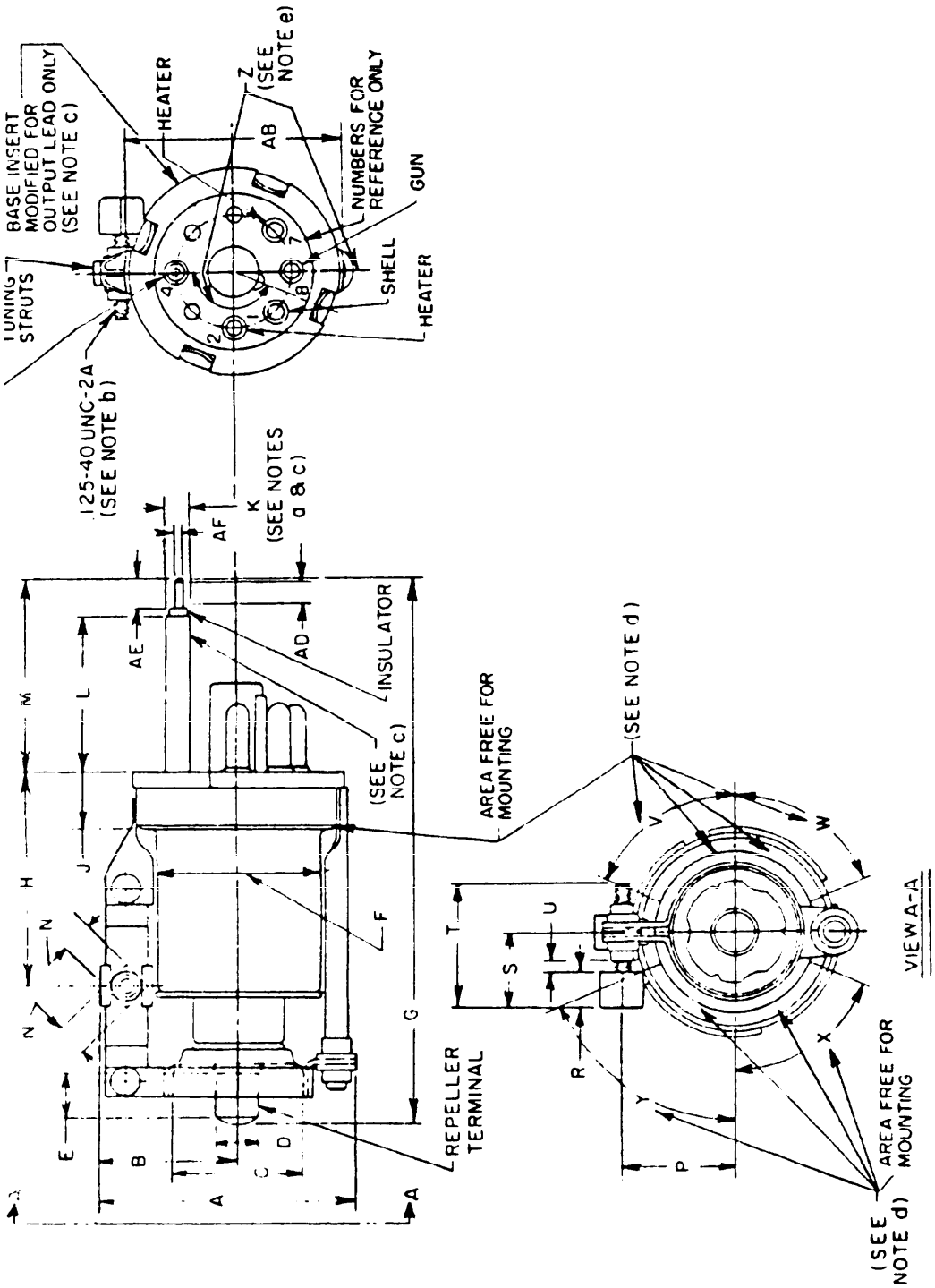


FIGURE 1. Outline drawing of electron tube type 6115A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:43Z
Check the source to verify that this is the current version before use.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses. (see note f)			
	Minimum		Maximum	
	Inches	mm	Inches	mm
Ⓓ	Qualification inspection			
C	.767	19.48	.797	20.24
T	.804	20.42	.835	21.21
V	---	---	70°	
W	---	---	70°	
X	---	---	70°	
Y	---	---	70°	
	Quality conformance inspection, part 1 (see note 5)			
H	1.32	33.53	1.44	36.58
J	.330	8.38	.400	10.16
K	See note b			
L	.88	22.35	.98	24.89
M	1.20	30.48	1.23	31.24
N	.182	4.62	.192	4.88
P	.66	16.76	.72	18.29
Z	154° 30'		160° 30'	
AF	.035	.89	.045	1.14
	Quality conformance inspection, part 2			
E	.240	6.10	---	---
F	1.00	25.40	1.02	25.91
R	.20	5.08	.24	6.10
S	.48	12.19	.52	13.21
U		---	.135	3.43
A	---	---	1.61	40.89
B	---	---	.86	21.84
D	.245	6.22	.255	6.48
G	3.25	82.55	3.50	88.90
AB	1.26	32.00	1.30	33.02
AE	.17	4.32	.20	5.08
	Reference dimension			
AD	.16 (4.06)			

NOTES:

a. The tuning screw shall be lubricated with oil-dag or equivalent noncorrosive lubricant. It shall be capable of being operated smoothly through its entire range without perceptible binding.

b. Nickel (30 M.S.I. silver plate permissible).

c. This base shall be capable of being inserted in a gage 1.22 (30.99 mm) thick having 4 holes .25 (6.35 mm) deep from the top of gage whose diameters are .103 (2.62 mm) for the contact pins. Remaining portion of hole to be clearance. Approximately .02 (.51 mm) larger in diameter and a fifth hole whose diameter .160 (4.06 mm) x 1.22 (30.99 mm) deep for the output lead. All holes located on the true centers. Also a center hole having the contour of the pilot but with a clearance of .002 (.05 mm) over the maximum diameter.

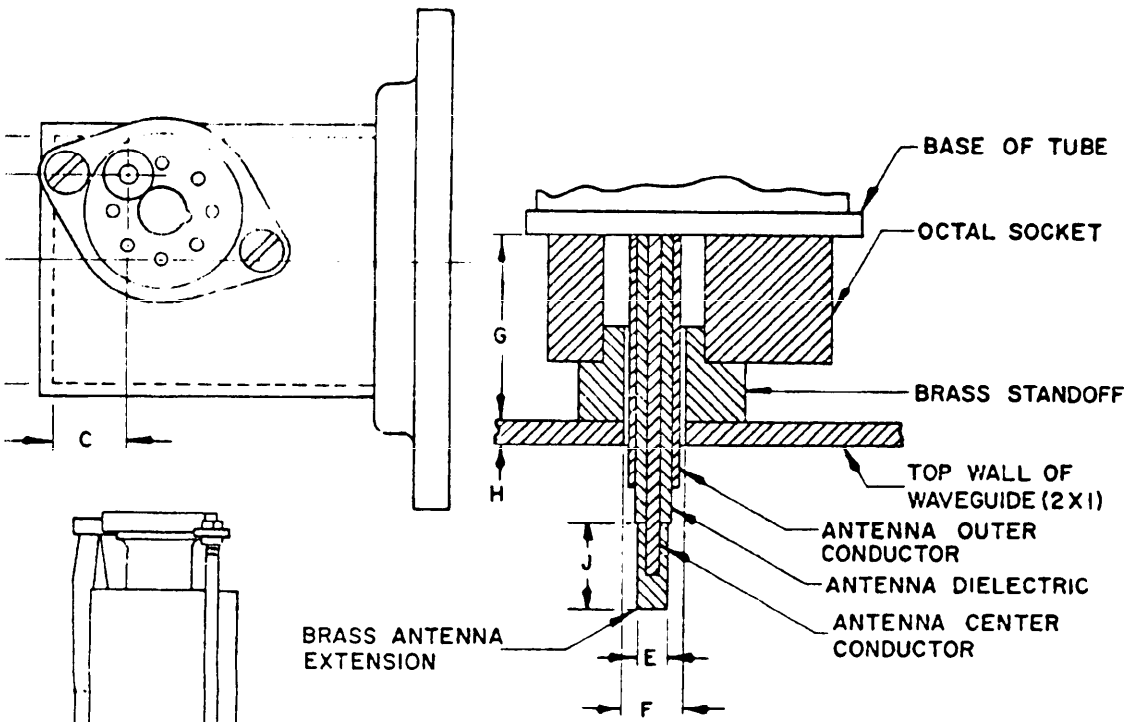
d. Applies to centerline passing through center of output lead and midway between tuning struts.

e. Shall be concentric with each other within .010 (.25 mm).

f. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Outline drawing of electron tube type 6115A - Continued.

Source: https://assist.dla.mil -- Downloaded: 202640-15T13:43Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (See note)	
	Minimum	Maximum
	Reference dimensions	
A	1.872 (47.55) Nom	
B	.62 (15.75) Nom	
C	.470 (11.94) Nom	
D	.872 (22.15) Nom	
E	.107 (2.72)	.113 (2.87)
F	.144 (3.66) Nom OD	
G	.814 (20.68)	.826(20.98)
H	.064 (1.63) Nom	
J	.328 (8.33)	.332 (8.43)

TE:
tric equivalents (to the nearest .01 mm) are given for general information only and are
d upon 1 inch = 25.4 mm.

FIGURE 2. Tube waveguide assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:43Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*.

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:43Z
Check the source to verify that this is the current version before use.

MENT OF THE NAVY
ELECTRONIC SYSTEMS COMMAND
GTON, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:43Z
Check the source to verify that this is the current version before use.